Roll No. .... Total Pages : 3

CMCA/M-22 **24528**

COMPUTER GRAPHICS AND ANIMATION

Paper–MCA-20-42

Time Allowed : 3 Hours] [Maximum Marks : 75

Note : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

Compulsory Question

1. Answer and **five** of the following questions in brief :

   (a) Why is Display Processor used in Graphics system?

   (b) How are colors obtained in Plasma Panel?

   (c) How are inputs given using a Touch Panel?

   (d) Write the equations for drawing a circle using polynomial method.

   (e) What will be the transformed position of a point (10, 14) if it is x-sheared by a factor of 3?

   (f) Write the equations for transforming a point (x_w, y_w) from window to viewport.

   (g) What is the role of Tweening in Animation?

UNIT–I

2. What is the role of Coordinate system in Computer Graphics? Describe any three important coordinate systems used in graphics and describe how they are used in creation and manipulation of pictures.

3. Answer the following questions in brief :

   (a) Describe the input device that is commonly used with a GUI based application.

   (b) What is the advantage of using look-up table?

   (c) How are pictures created in an LCD display?

UNIT–II

4. Describe one line scan converting algorithm in which the increment for x or y coordinate can be 1 or 0 only. Write the steps of the algorithm and use it to scan convert a line with endpoints (4, 5) and (11, 9).

5. (a) Describe how interpolation is used in drawing Bezier curves.

   (b) How are polygons filled using Boundary fill algorithm? How is Boundary fill algorithm different from Flood fill algorithm?

UNIT–III

6. Scale a rectangle with diagonal vertices at (4, 5) and (12, 13) to two times its original size keeping its centre fixed.

7. Distinguish between Cohen-Sutherland and Liang-Barsky line clipping algorithms with a description of the equations of line used by both.

UNIT–IV

8. Describe any one Euclidean geometry method and one Procedural method for 3-D solid object modeling. Also describe how a solid object can be shaded using Gouraud shading.

9. Distinguish between :

   (a) Parallel and Perspective projection.

   (b) Z-Buffer and Depth sorting hidden surface elimination methods.